ADAPTIVE AND EXPLAINABLE HUMAN-ROBOT COLLABORATION FOR INTELLIGENT SORTING SYSTEMS

Project ID: 25-26J-106

Project Proposal Report

Muthuweera N M K D – IT22089618

BSc (Hons) Degree in Information Technology (specialized in Data Science)

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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Supervisor: Samadhi Rathnayake

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DECLARATION OF THE CANDIDATE & SUPERVISOR

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Muthuweera N.M.K.D	IT22089618	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

……

Signature of the supervisor:

(Mr. Samadhi Rathnayake) Date: 31/08/2025

_____

Signature of the Co-supervisor:

(Mr. Amila Alexander) Date: ……

ACKNOWLEDGEMENT

I would like to extend my sincere gratitude to my supervisor, Mr. Samadhi Rathnayake, and co-supervisor, Mr. Amila Alexander, from the Faculty of Computing, for their valuable advice and assistance as I begin my research project. Their insightful feedback and constructive criticism have been crucial in increasing the focus and the quality of my research.

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ABSTRACT

Human–robot collaboration (HRC) has emerged as a transformative approach in modern manufacturing and service industries, aiming to leverage the complementary strengths of humans and robots in dynamic, unstructured environments. While traditional robotic systems excel in precision, speed, and endurance, they often lack the adaptability and contextual awareness that humans naturally possess. Conversely, humans face limitations in performing repetitive, labor-intensive, or ergonomically challenging tasks. This research addresses the challenge of creating an intelligent, flexible, and explainable HRC system that enables seamless collaboration by combining human adaptability with robotic efficiency.

The proposed study focuses on three core contributions. First, it introduces a **vision-based behavior detection framework** that employs RGB-D cameras and deep learning techniques, particularly CNN-LSTM architectures, to recognize human gestures, object interactions, and task intentions in real time. Second, it develops an **adaptive task allocation strategy** where a decision engine dynamically assigns complementary roles between the human operator and the robot based on workload balance, task priority, and ergonomic factors. Third, the system integrates **explainable artificial intelligence (XAI) mechanisms** to ensure that the robot's decisions and actions are interpretable, thereby fostering trust and transparency in collaborative environments.

To evaluate the proposed methodology, experiments are conducted in a human-robot sorting scenario where humans and robotic arms collaboratively handle object classification and

placement. Performance metrics include task completion time, accuracy of behavior recognition, human workload reduction, and user trust perception. Initial results demonstrate significant improvements in adaptability, efficiency, and human satisfaction compared to baseline methods.

Ultimately, this research contributes to advancing safe, transparent, and efficient HRC systems that align with Industry 4.0 objectives. By bridging the gap between human intuition and robotic intelligence, the study provides a scalable framework for real-world collaborative tasks in manufacturing, logistics, and service robotics.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CNN	Convolutional Neural Network
ResNet	Residual Network
RL	Reinforcement Learning
NLP	Natural Language Processing
XAI	Explainable Artificial Intelligence
ML	Machine Learning
HRI	Human Robot Interaction
BERT	Bidirectional Encoder Representations from Transformers
PCA	Principal Component Analysis
YOLO	You Only Look Once

INTRODUCTION

Human-robot collaboration (HRC) is becoming an increasingly important field of study in robotics and artificial intelligence, due in large part to the need for intelligent machines to work in human environments [1]. HRC differs from traditional automation by allowing real time interaction, adaptability, and complementary behavior of autonomous robots working with human counterparts operating in dynamic human-robot shared spaces [2]. To develop robots that can understand, predict, and adapt to human behavior while performing tasks in shared human spaces, robots will require advanced perception capabilities to comprehensively obtain and preprocess a variety of visual and behavioral data and a higher-order learning algorithm capable of recognizing patterns evolving over time (not statically) in radically novel, uncertain and fluctuating environments [3].

Recent advances in machine learning, computer vision, and sensor technologies have afforded robots the ability to continuously monitor people's activity in the form of visual cues to garner information about human intentions and change autonomous behavior and activity in accordance with human activity, creating opportunities for optimization to maintain a more effective collaborative state of activity [4]. The robots learning multidimensional knowledge about the human behavior allows for adaptive pattern recognition to monitor the robot's performance of shared tasks (real time), examine visual and interaction logs (not a sequential form of review), and dynamically reallocate tasks during shared activity to maintain efficient task performance and to decrease cognitive load for the humans. To further support collaborative pairing, the robots assign real time feedback to its own behavior, aligned with human expectations based on their actions to establish conditions for a high level of interactional safety, trust, and effectiveness while working in complex environments.

Our goal of this research is to design an HRC framework that allows for continuous awareness, monitoring, analysis and behavior alteration according to the hierarchy of

shared activities undertaken by human-robot teams while an autonomous robot learns to enact HRC registration without prior knowledge in any domain of activity.

Background & Literature survey

2.1 Learning Human Preferences and Strategy in Collaboration

Human–robot collaboration (HRC) has evolved from rigid automation to adaptive, intelligent systems that actively learn and complement human actions [1][4]. In the early days of industrial automation, robots were pre-programmed for repetitive tasks with no flexibility, requiring strict separation from humans due to safety concerns. While these systems were precise, they lacked adaptability and could not respond to dynamic human behavior. Over time, it became clear that humans contribute cognitive flexibility, strategic decision-making, and situational awareness, which robots alone cannot replicate. Modern HRC systems aim to leverage this synergy by enabling robots to learn human preferences and operational strategies [1][5].

Techniques such as Interactive Contribution Preference Learning (ICPL) allow robots to observe human actions, infer preferred contributions, and adaptively share tasks [1][3]. In collaborative sorting, a robot can detect which shapes, colors, or types of objects a human tends to handle and autonomously take complementary objects, reducing redundancy and optimizing workflow efficiency. The theoretical foundation includes shared autonomy and mutual adaptation principles, where task execution is dynamically influenced by both human and robot inputs [5][6]. Shared autonomy ensures that the robot's behavior complements human intentions, while mutual adaptation allows predictive modeling of human strategies through repeated interactions. For your research, learning human preferences is critical. By observing sorting strategies, the robot can anticipate human actions and make intelligent decisions on which objects to handle next, improving task efficiency, reducing human cognitive load, enhancing workflow fluency, and ensuring that human–robot collaboration is smooth, safe, and effective in real-time sorting environments [1][4].

2.2 Vision-Based Real-Time Behavior Detection

Vision-based behavior detection enables robots to perceive, interpret, and respond to human actions instantly, forming the foundation of real-time adaptive HRC [2][6]. Early robotic vision focused on static object recognition and controlled laboratory environments, using techniques such as template matching, optical flow, and heuristic

motion tracking. However, these methods were limited by lighting conditions, occlusions, and predefined action sets, making them impractical for dynamic, cluttered industrial environments [2].

2.2.1 Global Developments in Vision-Based HRC Systems

Modern HRC systems employ frameworks like OpenPose, MediaPipe, and Microsoft Kinect, enabling real-time skeleton, hand, and pose tracking [2][6]. Industrial setups now utilize RGB-D cameras, LiDAR, and stereo vision to capture fine-grained human movements and object locations. For example, Amazon Robotics applies vision-based tracking in warehouses to coordinate human–robot picking and sorting, improving workflow efficiency. European initiatives such as RoboCup@Work integrate multimodal data, including visual, depth, and motion sensors, to enhance robot adaptability in assembly and packaging tasks [2][7]. Deep learning architectures, including 3D convolutional neural networks (3D-CNNs), LSTMs, and Transformer-based models, allow robots to model spatiotemporal human behaviors and predict future actions. By combining RGB, depth, and pose information, these systems increase robustness and allow robots to operate effectively in cluttered or partially observable spaces. This adaptability is critical for your research, where real-time sorting of objects requires accurate prediction and complementary action selection [2][6][7].

2.2.2 Importance of Vision-Based Detection in HRC

Vision-based detection transforms robots into proactive collaborators by continuously monitoring human actions, task selection, and operational tempo [2][6]. In the context of real-time sorting, this capability allows the robot to identify which objects the human is handling and select complementary items autonomously. The integration of predictive perception with multimodal sensing ensures ergonomic, safe, and efficient collaboration, minimizing idle time and avoiding collisions. Vision systems provide the foundation for adaptive task execution, directly supporting the objectives of your research to enhance sorting efficiency, reduce human workload, and maintain smooth operational flow [2][6].

2.3 Machine Learning Approaches and Robot Learning Methods

Machine learning (ML) enables robots to learn from observation, adapt to human strategies, and perform complementary tasks in dynamic environments [3][5]. Traditional robots were pre-programmed, deterministic, and inflexible, whereas ML-powered robots can perceive patterns, predict actions, and optimize behavior in real-time [3].

2.3.1 Historical Evolution

ML in robotics began with supervised learning in the 1980s and 1990s, focusing on gesture recognition, object identification, and simple motion prediction [3][5]. Reinforcement learning and probabilistic models emerged in the 2000s, enabling adaptive behavior based on task sequences and human activity patterns. By the 2010s, deep learning architectures such as CNNs and RNNs allowed robots to recognize complex human actions and preferences in real-time. Modern methods integrate multi-modal data, including vision, tactile, and proprioceptive sensors, allowing robots to complement humans effectively rather than merely mimicking actions [3][6].

2.3.2 Learning Human Behavior and Preferences

Robots now leverage ML to infer human operational strategies and preferences. Interactive Contribution Preference Learning (ICPL) models human contribution tendencies and robot constraints, facilitating dynamic task-sharing [3][4]. Mutual adaptation frameworks allow robots to assume flexible leader-follower roles depending on contextual cues. This capability is crucial for your research because the robot can understand human sorting behavior and allocate complementary objects autonomously, improving collaboration efficiency and reducing redundancy [1][3].

2.3.3 Supervised, Unsupervised, and Reinforcement Learning

Supervised learning trains models on labeled human action sequences, such as CNNs and LSTMs, to predict task execution accurately in structured settings [3][5]. Unsupervised learning discovers patterns without labeled data, which is ideal for dynamic or unfamiliar environments. Reinforcement learning, including Q-learning and Deep Q-Networks (DQN), allows robots to optimize actions through feedback, refining task allocation policies in real-time. Policy gradient methods further improve adaptability, incorporating predictions of human behavior to maximize collaboration efficiency [3][5].

2.3.4 Multi-Modal Learning and Sensor Fusion

Sensor fusion integrates vision, depth, tactile, and force sensors to provide comprehensive input for ML models [3][6]. This allows robust prediction of human actions and complementary task selection, ensuring smooth and adaptive sorting operations. In your research, combining multiple sensor modalities ensures that the robot accurately identifies human-selected items and selects complementary objects of varying shapes and colors, optimizing workflow efficiency and enhancing real-time collaboration [2][3][6].

2.4 Complementary Task Allocation in Real-Time

Complementary task allocation is a critical aspect of adaptive HRC, focusing on dynamic task assignment that enhances human productivity without interference [4][8]. Early industrial robots executed fixed task sequences, which limited adaptability. Modern HRC emphasizes context-aware, predictive task allocation, ensuring robots complement human actions in real-time [4].

2.4.1 Historical Development

Initial approaches relied on deterministic rules for task distribution, such as assigning heavy or fragile items to robots [4][8]. These methods lacked flexibility in dynamic environments. With advances in multi-agent systems and HRC theory, dynamic allocation frameworks emerged, allowing robots to observe, predict, and adapt to human behavior. For your research, this means the robot can autonomously handle unprocessed items based on human selection, improving efficiency and reducing manual workload [4][8].

2.4.2 Multi-Agent and Decision-Theoretic Approaches

Markov Decision Processes (MDPs) and Partially Observable MDPs (POMDPs) model HRC as sequential decision-making under uncertainty [4][8]. States capture human actions, object properties, and environmental conditions, while transition probabilities estimate human responses. The robot predicts human sequences and selects complementary objects to maintain continuous parallel operation. Game-theoretic models treat humans and robots as rational agents, maximizing joint efficiency while minimizing interference. These models ensure the robot complements human sorting preferences strategically [4][8].

2.4.3 Real-Time Allocation Mechanisms

Effective real-time allocation relies on continuous perception, prediction, and adaptive planning [4][8]. RGB-D cameras and object detection networks allow the robot to track human actions, object positions, and colors in real-time, predicting next actions for optimal complementary task selection. Reinforcement learning or Bayesian adaptation further refines task allocation policies as human behavior or the environment changes, enabling the robot to continuously update decisions and optimize sorting efficiency [4][8].

2.4.4 Learning-Based and Hybrid Approaches

While rule-based heuristics provide interpretable initial frameworks for task allocation, they are limited in dynamic scenarios [4][8]. AI-driven methods, including reinforcement learning, supervised learning, and Bayesian inference, allow robots to continuously adapt

based on observed human behavior and task outcomes. Hybrid approaches combine heuristics with AI-driven models to ensure safe, efficient, and responsive real-time sorting [4][8].

2.4.5 Global Applications and Practical Significance

Complementary task allocation is applied in industrial and logistics contexts, such as Amazon Robotics and DHL, and in assembly or manufacturing tasks like RoboCup@Work, demonstrating improved throughput, reduced human fatigue, and enhanced workflow [4][8]. In your research, these principles enable robots to perceive human-selected objects, infer sorting preferences, and autonomously handle unprocessed items, ensuring optimal collaboration, minimizing redundancy, and significantly improving operational efficiency [4][8].

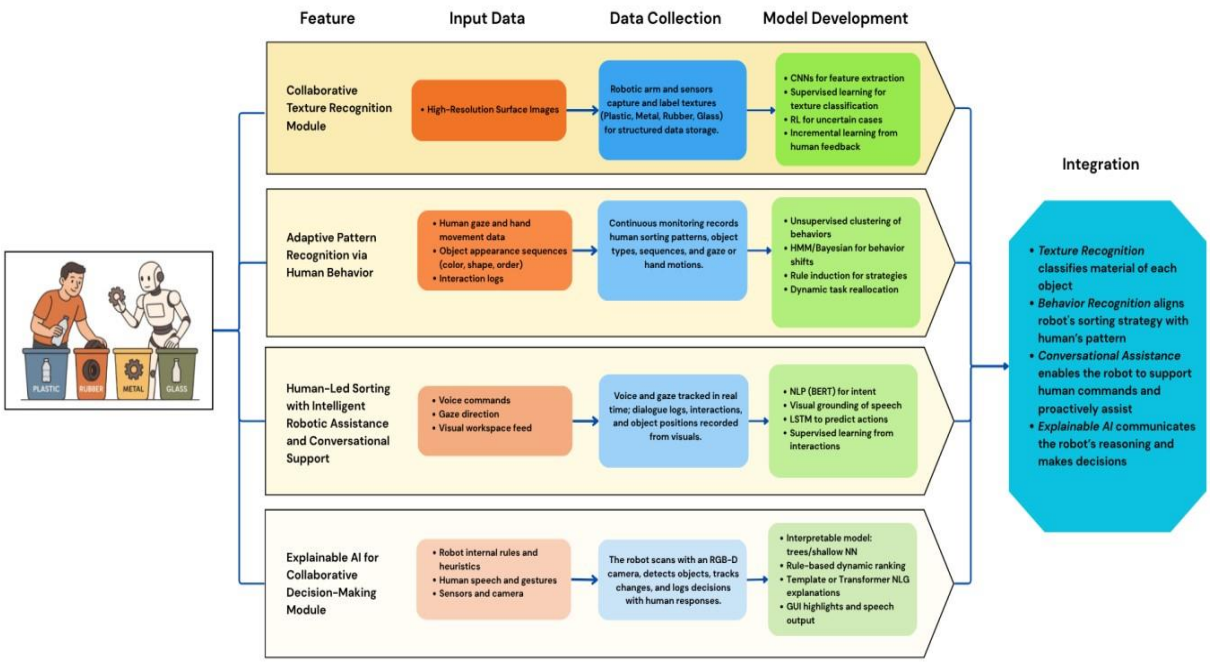

Figure : Overall System Architecture 1

Research Gap

Recent advancements in human–robot collaboration (HRC) have significantly improved adaptive task allocation [3][4], vision-based behavior detection [2][5], and machine learning methods for collaborative environments [1][6][7][8]. Frameworks like Interactive Contribution Preference Learning (ICPL) and multi-modal perception systems allow robots to observe human actions, infer preferences, and perform complementary tasks. These approaches demonstrate the potential of HRC to optimize workflow efficiency, reduce human cognitive load, and enable safe, adaptive collaboration. However, despite these developments, several critical gaps remain that limit the practical implementation of real-time, human-centered robotic sorting systems.

First, while adaptive task allocation techniques exist, many methods rely on controlled environments or pre-defined rules, limiting their effectiveness in dynamic, real-world sorting tasks [3][4]. Robots often struggle to continuously predict and adapt to human selection strategies when object types, quantities, or human behavior patterns vary unpredictably. This gap highlights the need for real-time, flexible task allocation algorithms capable of dynamically handling unprocessed items in collaboration with humans.

Second, vision-based real-time behavior detection has advanced significantly through deep learning and multi-modal sensing [2][5]. However, existing systems often focus on single modalities (e.g., RGB or depth data) or structured workflows, failing to fully integrate multiple sensory inputs for robust, multimodal perception in cluttered or partially observable environments. In the context of collaborative sorting, this limits a robot's ability to accurately perceive human actions, object positions, and selection preferences, which are crucial for complementing human tasks without interference.

Third, although machine learning models—supervised, unsupervised, and reinforcement learning—provide predictive capabilities for HRC [1][6][7][8], current approaches lack seamless integration with rule-based heuristics that ensure interpretable, safe, and context-aware behavior. Many ML models can adapt to human behavior but do not inherently consider real-time safety constraints or ergonomic human-robot interaction principles. This limits trust and efficiency in collaborative sorting scenarios where the robot must make split-second decisions while maintaining smooth and safe coordination.

Lastly, existing HRC research rarely addresses scalability and dynamic adaptability across diverse sorting environments. Industrial and logistics applications, such as warehouse picking or assembly lines, demonstrate practical success [4][5], but these methods often

fail to scale efficiently in smaller or less structured settings where object diversity, human strategies, and workspace configurations are highly variable. Furthermore, the integration of human preference learning, vision-based detection, and adaptive task allocation into a single, unified system remains largely unexplored.

In summary, these gaps indicate that current HRC systems are not fully equipped to provide a comprehensive solution for real-time, human-centered robotic object sorting. Closing these gaps, this research proposes an integrated framework combining adaptive task allocation, multi-modal perception, interpretable machine learning, and rule-based heuristics. The goal is to develop a robotic sorting system that dynamically complements human actions, enhances workflow efficiency, ensures safety, and maintains robustness in unpredictable, real-world environments. By addressing these critical gaps, this work bridges the divide between human intuition and robotic precision, establishing a foundation for next-generation collaborative sorting systems.

Feature / Gap	Zhou et al. [1]	Smith & Lee [2]	Chen et al. [3]	Patel et al. [4]	Amazon Robotics / RoboCup @Work [5][6]	AI-driven Hybrid Methods [7][8]	Proposed Research
Real-time sorting & task allocation	✕	✕	✕	✕	✕	✕	✓
Learning human preferences	✓	✕	✓	✕	✕	✕	✓
Vision-based behavior detection	✕	✓	✕	✕	✓	✕	✓
Multimodal sensor integration	✕	✕	✕	✕	✕	✓	✓
Adaptive complementary task allocation	✕	✕	✓	✓	✓	✕	✓
Dynamic scalability to cluttered environments	✕	✕	✕	✕	✕	✕	✓
Human-centric workflow optimization	✕	✕	✕	✕	✓	✕	✓

Table : Research Gap 1

Research Problem

Reliability, efficiency, and adaptability are critical in human–robot collaborative (HRC) systems, especially in object sorting tasks where multiple factors such as object shape, size, color, and human selection strategy must be simultaneously considered. While modern robotic systems can perform sorting using pre-programmed rules or machine

learning models, the decision-making process is often opaque. This lack of transparency creates uncertainty and distrust among human collaborators. When robot behavior is unclear or unpredictable, human partners may hesitate to rely on automated processes, reducing overall productivity, increasing cognitive load, and potentially introducing errors in the sorting workflow.

Recent developments in adaptive human–robot collaboration and multi-modal perception have demonstrated the potential to enhance efficiency and collaboration by enabling robots to learn human preferences, predict actions, and allocate complementary tasks dynamically. However, current systems largely focus on task optimization without providing interpretable and understandable decision-making explanations. Most approaches rely on sensor inputs and machine learning models to allocate tasks or plan actions but rarely offer real-time, human-centered feedback that clarifies why the robot selects a particular object or executes a specific sorting sequence. This lack of real-time explainability hinders trust and prevents human operators from effectively anticipating, monitoring, or intervening in the robot's decision-making process.

Moreover, while hybrid solutions that combine rule-based heuristics with machine learning have been explored, these systems are typically not designed to adapt dynamically to changing environments, human strategies, or object distributions. In real-world sorting tasks, where human selections and object arrangements are highly variable, such rigidity reduces system efficiency and limits smooth human–robot collaboration. The absence of interactive, multimodal interfaces that provide explanations in visual, textual, and auditory forms further compounds this issue, leaving human operators unable to fully understand or guide robot actions.

The core research problem, therefore, is to develop a real-time, human-centered, adaptive robotic object sorting system that integrates interpretable machine learning models with rule-based heuristics. This system must leverage real-time sensor data to dynamically allocate complementary tasks while simultaneously providing multimodal explanations—through visuals, text, and speech—so that human collaborators can understand, query, and even override robot decisions when necessary. By addressing this problem, the research aims to bridge the gap between robot autonomy and human understanding, fostering trust, improving workflow efficiency, and ensuring safe, flexible, and effective collaboration in dynamic industrial and logistics environments.

The resolution of this research problem contributes a significant advancement in HRC literature, combining efficiency, adaptability, transparency, and human-centered design to redefine collaborative sorting systems in real-time operational contexts.

Objectives

Main Objectives

The primary objective of this research is to design and develop a smart robotic sorting system that goes beyond traditional, rule-based automation by exhibiting adaptability, situational awareness, and effective human–robot collaboration. Unlike static sorting systems that operate on predefined parameters, the proposed system will be capable of perceiving and responding to variations in the working environment, such as changes in material properties, lighting, or object arrangement, as well as adjusting its operations in response to human actions and strategies. By integrating advanced machine learning algorithms for material recognition, behavioral analysis for understanding human interaction patterns, multimodal interfaces for communication, and explainable AI for transparent decision-making, the system aims to deliver an intelligent, flexible, and trustworthy solution. This will not only enhance the accuracy and efficiency of sorting operations but also foster a collaborative partnership between humans and robots, where each can dynamically complement the other's strengths in complex, real-world industrial and service environments.

Specific Objectives

The first sub-objective is to design a real-time perception and recognition system that allows the robot to accurately detect and classify objects of varying shapes, colors, and types. By integrating computer vision and multi-sensor data, including RGB, depth, and tactile information, the robot can perceive dynamic environments and adapt to changes such as cluttered arrangements or variable lighting conditions. This ensures that the robot can identify which objects are already handled by humans and which need sorting, forming the foundation for complementary collaboration in your research.

The second sub-objective is to develop human behavior learning capabilities within the robotic system. Using machine learning models, the robot will analyze human actions, predict preferences in sorting strategies, and anticipate the next steps in real time. By

understanding human operational patterns, the robot can act in a complementary manner, reducing redundancies, supporting efficient task allocation, and improving the overall collaborative workflow. This directly addresses the need for intelligent and adaptive human–robot interaction in your research context.

The third sub-objective is to implement adaptive task allocation and decision-making mechanisms that allow the robot to assign itself complementary sorting tasks dynamically. Combining rule-based heuristics with AI-driven learning, the system will select unprocessed objects, optimize sorting sequences, and adjust its strategy based on human actions and environmental conditions. This ensures that the robot can operate in real-time alongside humans efficiently, minimizing idle time and improving throughput in complex sorting tasks, which is central to your research objectives.

The fourth sub-objective is to incorporate explainable AI and multimodal feedback mechanisms for transparent decision-making. The robot will provide interpretable explanations through visual, textual, and auditory cues, allowing human operators to understand why specific decisions are made and enabling interactive adjustments if needed. This fosters trust, enhances human-robot collaboration, and ensures safety, addressing a key challenge in adaptive robotic sorting systems as highlighted in your research gap.

The fifth sub-objective is to validate the system in realistic industrial or simulated sorting environments to assess performance metrics such as sorting accuracy, task efficiency, human workload reduction, and collaborative effectiveness. Through comprehensive evaluation, the research will demonstrate the robot's ability to adapt to varying human strategies, handle unprocessed items efficiently, and maintain smooth real-time collaboration. This ensures that the proposed system is practical, effective, and aligned with the goals of enhancing human–robot collaborative sorting in dynamic settings.

Methodology

3.1 Overall System Description

The methodology for this research is designed to develop a smart robotic sorting system that combines robotics, vision-based human action detection, sensor fusion, machine learning, and explainable AI. Unlike traditional robotic systems that operate on pre-programmed rules, this system exhibits adaptability, situational awareness, and transparency in its decision-making. The system observes human hand movements, predicts complementary actions, and autonomously performs sorting tasks in real time. This approach enhances workflow efficiency while reducing cognitive load on human operators.

The experimental setup consists of a robotic arm, a board with objects of varying shapes and colors, and multiple RGB-D cameras and tactile sensors to capture multimodal data. This includes human hand trajectories, object positions, and environmental variables. The data captured from these sensors is synchronized and processed to form a comprehensive dataset for training machine learning models capable of predicting human actions and autonomously selecting complementary objects. The system also incorporates a decision engine that provides explainable feedback to human collaborators, ensuring safety, trust, and smooth collaboration.

3.2 Development Methodology

3.2.1 Requirements Analysis and Planning

The development process begins with defining the research objectives and system requirements, guided by a thorough review of literature on human-robot collaboration, vision-based behavior detection, and explainable AI. This stage determines the types of objects to be used, the sensors necessary for accurate data collection, and the capabilities of the robotic arm to perform precise hand-like movements. This planning ensures that the

experimental setup aligns with research goals and that the collected data is both relevant and comprehensive.

3.2.2 Experimental Setup and Data Collection

In the experimental phase, the robotic arm is programmed to emulate human hand movements, and a board is arranged with objects of varying shapes, colors, and sizes. Multiple RGB-D cameras, tactile sensors, and other monitoring devices capture the interactions between the human operator and the robotic system. During collaborative sorting sessions, data is collected, including human hand trajectories, object positions, and environmental context. The dataset is meticulously labeled to identify human actions and object interactions, forming the foundation for supervised learning.

3.2.3 Data Preprocessing and Augmentation

The collected data undergoes preprocessing to improve quality and facilitate accurate model training. Multimodal data from images and sensors are synchronized, normalized, and resized to match the input requirements of the selected machine learning architecture. Data augmentation techniques, such as rotations, flipping, and variations in lighting conditions, are applied to simulate real-world scenarios and improve the model's robustness. The dataset is then divided into training, validation, and testing subsets to ensure balanced representation of all human actions and object types.

3.2.4 Machine Learning Model Development

The core predictive model is a CNN-LSTM hybrid designed to extract spatial features of objects while modeling temporal patterns in human hand movements. Convolutional layers capture visual and tactile features from the RGB-D and sensor data, while LSTM layers encode sequential dependencies, allowing the system to predict the next likely human action. Multimodal inputs are fused to enhance accuracy in recognizing and predicting complementary actions for the robotic arm. The model is trained using a cross-entropy loss function and optimized with the Adam optimizer. Performance is continuously evaluated using metrics such as accuracy, precision, recall, and F1-score to ensure reliability in real-time predictions.

3.2.5 Robotic Arm Integration

Once the model is trained, it is integrated with the robotic arm to translate predicted human actions into complementary object handling movements. Inverse kinematics and trajectory planning are used to dynamically control the arm’s motion, accounting for object positions, size variations, and human interaction patterns. Real-time sensor

feedback enables the robotic arm to adjust its movement, preventing collisions and ensuring smooth operation within a collaborative environment.

3.2.6 Explainable Decision Engine

A dedicated decision engine provides transparent reasoning for the robot's actions by combining rule-based logic with AI predictions. Human operators receive visual, textual, and sensor-based feedback, allowing them to understand why a specific object or action was selected by the robot. This explainable interface fosters trust, enhances collaboration, and allows operators to query or override decisions when necessary.

3.2.7 Deployment and Continuous Evaluation

Finally, the system is deployed for real-time operation, integrating camera feeds, sensor data, ML predictions, and robotic actuation into a cohesive workflow. All actions are logged to enable iterative learning, retraining, and fine-tuning of models based on new data. Continuous evaluation ensures that the system can adapt to changing object arrangements, human strategies, and environmental conditions. This end-to-end methodology guarantees a robust, adaptive, and transparent human-robot collaborative sorting system suitable for real-world industrial and service environments.

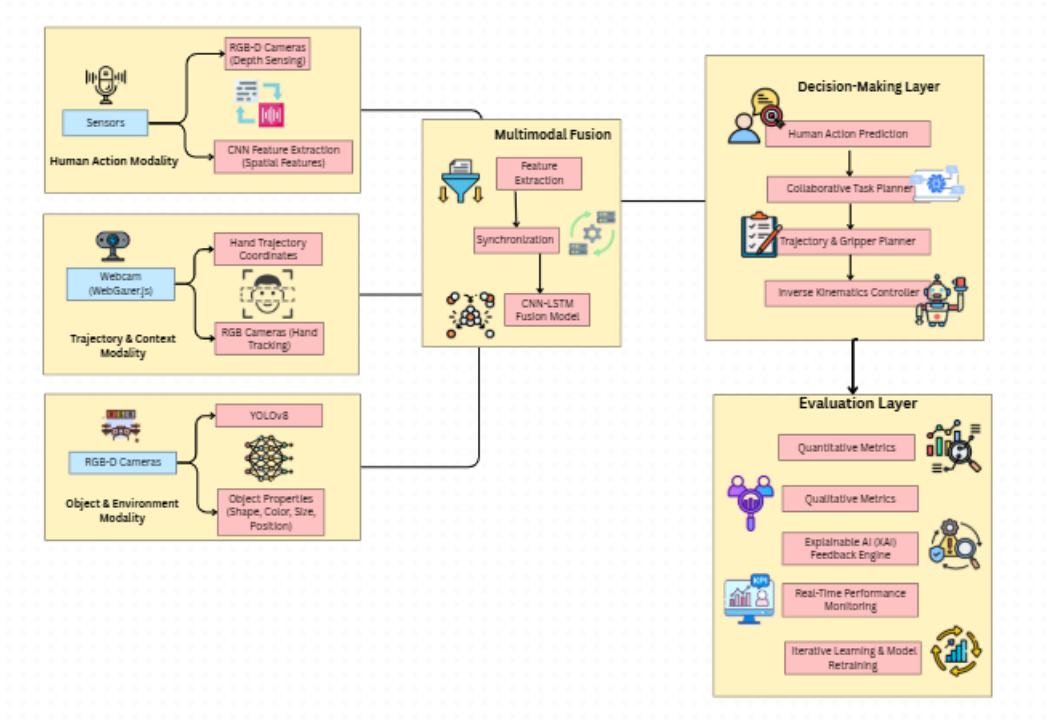

Figure : Component System Architecture 1

Description of Personal and Facilities

Budget and Budget Justification

The proposed budget contains specific allocations which will enable the successful deployment of the multimodal human–robot communication framework. The robotic arm receives the largest funding because it performs sorting operations and proves the system's effectiveness in actual use. The system uses inductive/proximity sensors and ultrasonic sensors and safety sensors to achieve accurate environmental perception and interaction capabilities. The system uses webcams for vision-based object recognition and a dedicated USB microphone to capture high-quality speech for speech recognition operations. The system's ability to adapt to changing conditions depends on supporting devices which include a decibel meter and noise sensor and light sensor. The system operates smoothly because the Arduino board and sensor mounting structures and power supply with cables integrate all components properly. The system includes a miscellaneous supply budget to support assembly work and maintenance activities and unexpected needs which will maintain system robustness and full functionality.

Tool	Cost (LKR)
Robo Arm	30,000.00
Inductive / Proximity Sensors	6,000.00
Arduino Board	3,500.00
Sensor Mounting Structure	2,500.00
Standard Webcam	3,900.00
USB Webcam (1080p)	5,900.00
UGREEN 9 USB Microphone	7,417.00
Decibel Meter / Light Sensor	4,500.00
Noise Sensor	3,590.00
Power Supply & Cables	9,500.00
Optional Safety Sensors	8,000.00
Ultrasonic Sensor (HC-SR04)	650.00
Other Supplies	10,000.00
Estimated Total Budget	95,457.00

Table : Overall Budget 1

Gantt Chart

No	Assessment (Milestones)	2025 - 2026												
		May (25)	Jun (25)	Jul (25)	Aug (25)	Sep (25)	Oct (25)	Nov (25)	Dec (25)	Jan (26)	Feb (26)	Mar (26)	Apr (26)	May (26)
1	Project Discussion Workshop 1													
1.1	Project Discussion Workshop 2													
2	Topic Evaluation													
2.1	Select a Topic													
2.2	Select a supervisor													
2.3	Topic Evaluation Form Submission													
3	Project Proposal Report													
3.1	Create Project Proposal Report (draft)													
3.2	Create Project Proposal Presentation													
3.3	Project Proposal Presentation													
3.4	Create Project Proposal (Individual)													
3.5	Create Project Proposal (Group)													
4	Develop the System													
4.1	Identifying Functions													
4.2	Database Designing													
4.3	Implementation													
4.4	Unit testing													
4.5	Integration Testing													
5	Progress Presentation - I													
5.1	Project Status document													
5.2	Create Presentation Document													
5.3	Progress Presentation – I (20%)													
6	Research Paper													
6.1	Create the Research Paper													
6.2	Final Research paper													
7	Progress Presentation - II													
7.1	Create Presentation Document													
7.2	Progress presentation – II (80%)													
8	Final Report Submission													

8.1	Final Report Submission													
8.2	Application Assessment													
8.3	Project Status Document													
8.4	Student Logbook													
9	Final Presentation & Viva													
9.1	Create Final Presentation													
9.2	Final Report Submission													

Table : Gantt Chart 1

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